Questions on BJT, 2019

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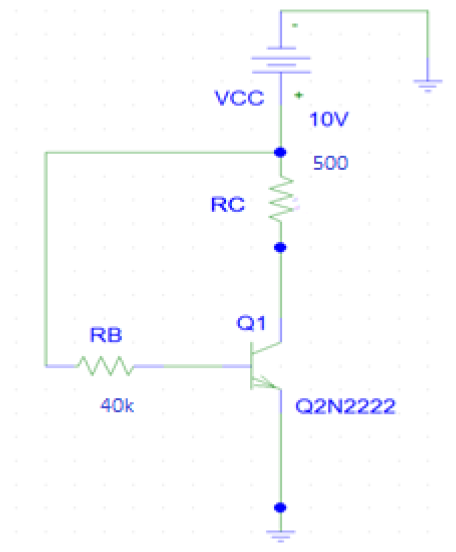
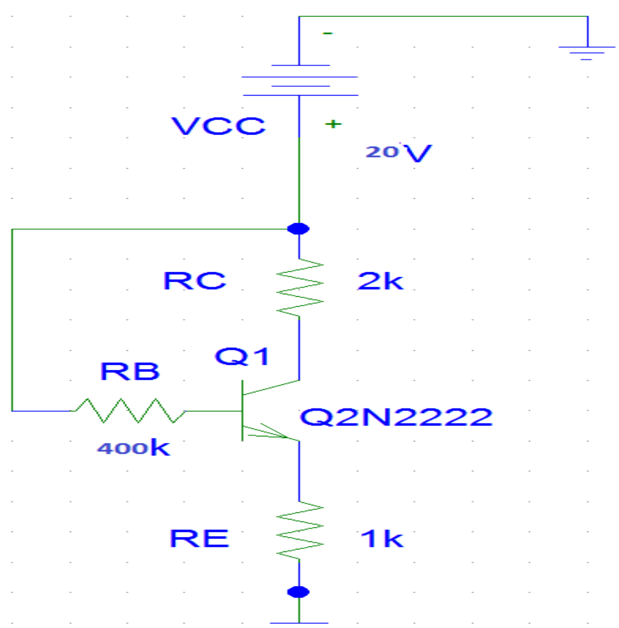
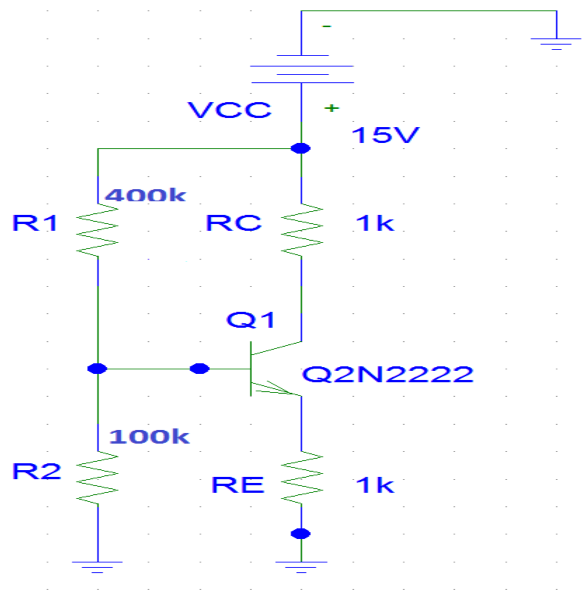
1. Draw the dc load line of the given circuit and calculate the Q-point values of I_C and V_{CE}. Consider $\beta_{dc} = 50$.



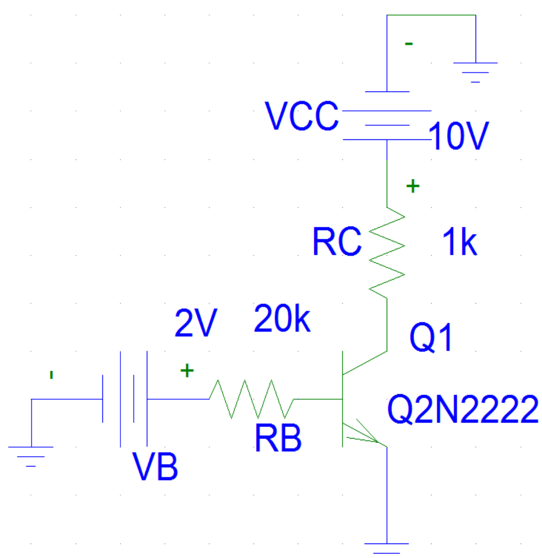
2. Draw the dc load line of the given circuit and calculate the Q-point values of I_C and V_{CE}. Consider $\beta_{dc} = 50$ and $I_C \neq I_E$.



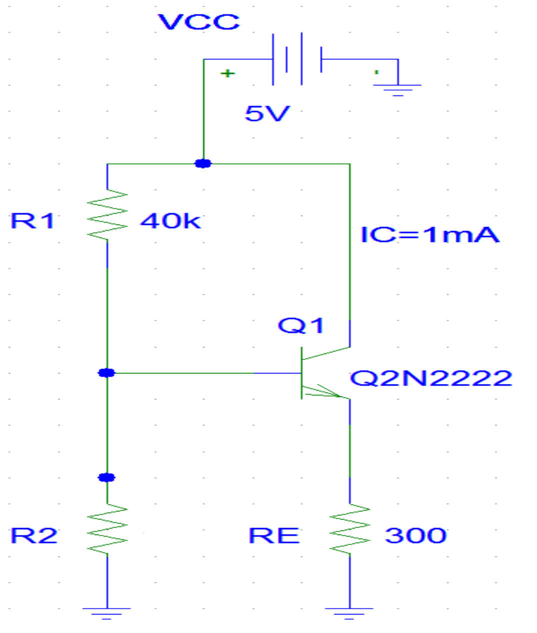
3. Determine I_C and V_{CE} using Thevenin's theorem in the circuit shown below. Assume β_{dc} = 99 and $I_C \neq I_E$.



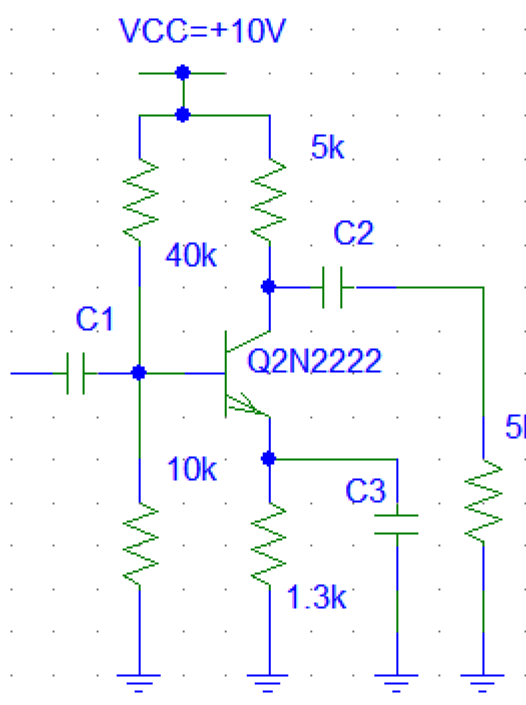
4. Determine the working region of the BJT in circuit below. Assume $\beta_{dc} = 100$.



5. Calculate the value of R_2 which results in I_C=1mA for the Si BJT as shown below. Assume that the BJT has a very high β_{dc}.

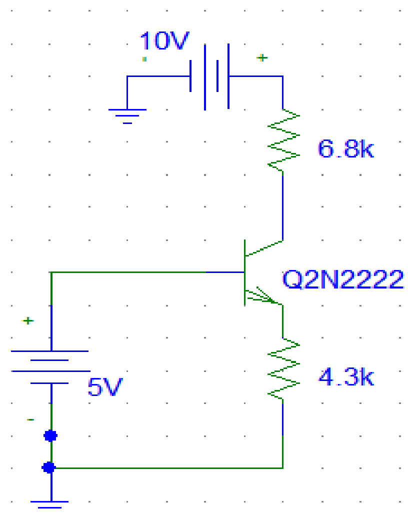


6. For the given circuit, V_{BE}=0.7V and β_{dc} is very large. Do the DC analysis and find the value of I_E.

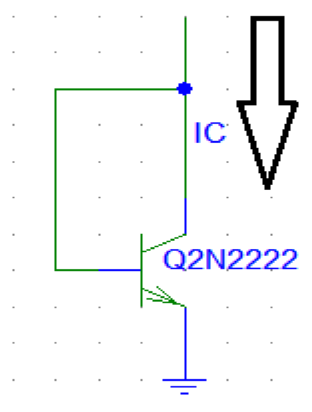


7. For problem 6, assume that V_T=25mV and all the capacitances are very large. Do the small signal analysis and calculate the mid-band voltage gain of the amplifier.

8. Assume, V_{BE}=0.7V and β_{dc} is very large and identify in which mode the BJT is working.



9. Assume, V_{BE}=0.7V and the reverse saturation current of the junction at room temperature is 10^{-14}A. Calculate the emitter current for η=1.



10. For the circuit shown in below, given that $V_{CE(sat)}$=0.2V, β_{dc}=200. Calculate the minimum base current required to drive the BJT in saturation.